

EDUCATIONAL ATTAINMENT AND SUICIDE RATES IN THE UNITED STATES¹

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Summary.—We examined the relationship between educational attainment and suicide rate in the United States for 2001. Suicide rates, adjusted for age, were compared with percentage of college graduates, median household income, and poverty in 50 states in 2001. The correlations of suicide rates with educational attainment and median household income were both negative and statistically significant. Poverty was not significantly related to suicide rates. We concluded that higher education and income were associated with a decrease in suicide rates in 2001. Data from other years require examination for this conclusion to be generalizable.

The role of educational attainment as a factor affecting suicide rates has been relatively unstudied. Stack (1998) reported that for every year of education attainment, suicide rates declined by 2% for Caucasian males. However, another study indicated a higher suicide rate among college students compared to less educated age-matched controls (Ross, 1969). Studies examining correlates of educational attainment such as intelligence and socioeconomic status (Neisser, Boodoo, Bouchard, Boykin, Brody, Ceci, Halpern, Loehlin, Perloff, Sternberg, & Urbina, 1996; Kirkcaldy, Furnham, & Siefen, 2004) yielded contradictory information. Whereas intelligence was negatively correlated with suicide rates in studies of men in the Australian and Swedish military (O'Toole & Cantor, 1995; Gunnell, Magnusson, & Rasmussen, 2005), a positive correlation was found among men in the Israeli military (Apter, Bleich, King, Kron, Fluch, Kolter, & Cohen, 1993). Other studies have reported either positive correlations between national IQ estimates and suicide rates (Lester, 1984, 1995, 2003; Voracek, 2004) or no significant association (Lester, 1993). With regard to socioeconomic status, although classic works on suicide contended correlation is positive between socioeconomic status and suicide, that is, the higher the socioeconomic status, the higher the suicide rate (Henry & Short, 1954; Durkheim, 1955/1966), several studies have instead reported negative correlations between socioeconomic status and suicide rates (Schony & Grausgruber, 1987; Boxer, Burnett, & Swanson, 1995). These disparate findings prompted us to conduct an additional study of the relationship between suicide and educational attainment taking into account household income and poverty for the 50 U.S. states.

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METHOD

Suicide rates for the year 2001 for each state, adjusted for age, were obtained from the National Center for Health Statistics (Arias, Anderson, Murphy, & Kochanek, 2003). The percentage of college graduates for the year 2001 per state was obtained from the U.S. National Center for Education Statistics (2001). Median household income and percent people below the national poverty line for the year 1995 in each state were obtained from the U.S. Bureau of the Census (1995). Suicide rates are age-adjusted and are a weighted average of suicide rates using rates weighted by the fixed population proportions by age. The data were analyzed by stepwise forward multiple regression.

RESULTS AND DISCUSSION

Suicide rates, adjusted for age, were negatively and significantly correlated with percentage of college graduates in the state ($r = -.39$, $p < .01$) and median household income ($r = -.33$, $p < .01$). The correlation between suicide rates and poverty was positive ($r = .20$) but not statistically significant. Stepwise multiple-regression analysis, using percentage of college graduates, median household income, and percent households below the national poverty level as predictor variables, was significant only for percentage of college graduates ($r^2 = 0.15$; Beta = -0.28 ; $F_{1,28} = 8.5$, $p < .005$).

These results are similar to other studies in which higher educational attainment has been associated with lower suicide rates (Ross, 1969; Stack, 1998) and with studies reporting similar negative correlations with variables associated with educational attainment such as intelligence (Apter, *et al.*, 1993; O'Toole & Cantor, 1995; Gunnell, *et al.*, 2005) and socioeconomic status (Schony & Grausgruber, 1987; Boxer, *et al.*, 1995) and are not consistent with reports of a positive correlation between international economic income derived from Lynn and Vanhanen (2002) and suicide rates (Lester, 1984; Voracek, 2004).

One reason for the different outcome between educational attainment and its proxies, especially national intelligence, is that the latter does not rely on standardized measures of intelligence, e.g., IQ tests, but on questionable indices assumed to be correlated with intelligence, e.g., gross national product, and often compares IQ scores of children and adults (Pontikos, 2005²). In critiquing the reported association between IQ and economic productivity, Volken (2003) added IQ alone does not lead to wealth but its conversion by training, i.e., educational attainment. Increasing education in fact raises IQ scores from 1.0 to 4.2 points for every additional year of schooling (Winship & Korenman, 1997). While there is a strong correlation between

²Pontikos, D. (2005) Greek IQ. <http://dienekes.angeltowns.net/articles/greekiq/>

educational attainment and intelligence, education's potential to affect IQ, and hence income, depends on pre-existing disparities in income between countries in which people can receive higher education. Translating national ratings of different IQ tests into standardized scores and using those to compare countries in terms of suicide rates is an interesting exercise but subject to question.

Our finding that other correlates of educational income, such as household income, were negatively associated with suicide rate has been frequently reported by others (Maris, Berman, & Silverman, 2000) and so is not surprising given the negative correlation between educational attainment and suicide. On the other hand, poverty was not significantly related to suicide rates.

We conclude that higher educational attainment was associated with a decrease in suicide rates in the United States in 2001. Additional studies for other years and other countries must be reported for this conclusion to be generalizable.

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